IRT-FIT: Detecting Benchmark Contamination in Large Language Models via Item Response Theory Person-Fit Statistics

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Abstract

Benchmark contamination undermines the reliability of Large Language Model (LLM) evaluation, yet all existing detection methods require privileged access—training data, token probabilities, or model weights. We introduce IRT-FIT, a contamination detection framework that operates at the *minimal-information operating point*, requiring only binary (correct/incorrect) response patterns. Drawing on Item Response Theory (IRT) person-fit statistics originally developed for detecting exam cheating, we fit a four-parameter logistic (4PL) model to the item–examinee response matrix and compute the standardized person-fit statistic l_z^*. In controlled contamination experiments across four model families and three benchmarks, IRT-FIT achieves AUROC 0.94 on MMLU at 15% contamination (Cohen's d=2.1). We characterize detection power as a predictable function of benchmark size (R^2=0.97 across seven benchmarks), with power exceeding 90% for benchmarks with $\geq$10K items at 15% contamination. Applying IRT-FIT to 11,200 models from the Open LLM Leaderboard, we identify systematic contamination patterns in 8.3% of submissions, revealing temporal contamination waves (+2.1 percentage points per quarter) and model-family clustering in 12 organizations. Critically, IRT-FIT retains 63% of its detection signal after reinforcement learning fine-tuning (AUROC 0.88), whereas token-probability methods collapse to near-chance (AUROC 0.58), directly addressing the post-RL detection failure identified by **?**.

1 Introduction

Benchmarks are the primary currency of LLM evaluation, yet their validity is increasingly threatened by contamination—the inclusion of benchmark items in training data. As training corpora scale to trillions of tokens, the probability of inadvertent or deliberate benchmark exposure grows, producing inflated scores that mislead practitioners and researchers alike.

Existing detection methods span a wide information-access spectrum. N-gram overlap methods require access to training data (**?**). Membership Inference Attacks (MIA) such as Min-K%++ (**?**) require token-level probabilities. Time Travel probing (**?**) requires crafted prompts and model interaction. ConStat (**?**) requires re-testing with rephrased items. Critically, **?** demonstrate that *all* MIA-based methods fail when models undergo post-training reinforcement learning (RL), as RL redistributes token probability mass while preserving behavioral contamination. No existing method operates at the minimal-information level—binary response patterns alone.

We observe that a contaminated LLM is functionally analogous to an examinee with item preknowledge: it produces correct responses on items beyond its estimated ability, generating aberrant response patterns detectable by person-fit statistics. Item Response Theory (IRT) provides a mature, well-characterized statistical framework for exactly this detection problem, developed over decades in educational testing (**??**). We repurpose this framework for LLM contamination detection, establishing the *minimal-information operating point* on the contamination detection frontier.

Our contributions are:

1. We introduce IRT-FIT, a contamination detection framework using IRT person-fit statistics (l_z^*) that requires *only binary response patterns*. Controlled experiments across 4 model families, 3

benchmarks, and 6 contamination levels demonstrate AUROC 0.94 on MMLU at 15% contamination with Cohen's d=2.1.

2. We characterize detection power as a predictable function of benchmark size, confirming power $\propto \sqrt{n_{\text{items}}}$ across 7 benchmarks (R^2=0.97). We identify two distinct signal regimes: partial contamination produces aberrant patterns ($l_z^* \ll 0$), while full contamination produces over-consistent patterns ($l_z^* \gg 0$).

3. We conduct the largest contamination audit to date, scanning 11,200 models from the Open LLM Leaderboard. We flag 8.3% of submissions with systematic contamination patterns, revealing temporal trends, model-family clustering, and organization-level contamination.

4. We demonstrate that IRT-FIT retains 63% of its detection signal post-RL (AUROC 0.88), while Min-K%++ collapses to AUROC 0.58, directly addressing the open problem of post-RL contamination detection.

2 Background

2.1 Item Response Theory

IRT models the probability that examinee j with latent ability θ_j responds correctly to item i with parameters (a_i, b_i, c_i, d_i). Under the four-parameter logistic (4PL) model:

$$P(X_{ij} = 1 \mid \theta_j) = c_i + (d_i - c_i) \cdot \frac{1}{1 + \exp(-a_i(\theta_j - b_i))} \tag{1}$$

where a_i is discrimination, b_i is difficulty, c_i is the guessing parameter (lower asymptote), and d_i is the upper asymptote (feasibility).

The 4PL model extends the standard 3PL by allowing $d_i < 1$, accommodating items where even high-ability examinees cannot achieve perfect accuracy—a common phenomenon in LLM evaluation due to format sensitivity and prompt artifacts.

2.2 Person-Fit Statistics

Person-fit statistics detect aberrant response patterns—responses inconsistent with the estimated ability under the assumed IRT model. The standardized l_z^* statistic (**??**) is defined as:

$$l_z^* = \frac{l_0 - E(l_0)}{\sqrt{\text{Var}(l_0)}} \tag{2}$$

where $l_0 = \sum_{i=1}^{n} [x_{ij} \ln P_i + (1 - x_{ij}) \ln(1 - P_i)]$ is the log-likelihood of the response pattern under the IRT model, and $P_i = P(X_{ij} = 1 \mid \hat{\theta}_j)$.

Under the null hypothesis (no aberrance), $l_z^* \sim N(0, 1)$ asymptotically. Extreme negative values indicate *aberrant* patterns (unexpected failures or unexpected successes), while extreme positive values indicate *over-consistent* patterns (responses too predictable under the model). In the contamination context: partial contamination yields $l_z^* \ll 0$ (the model "knows" answers to items beyond its ability), while full contamination yields $l_z^* \gg 0$ (near-perfect performance appears over-consistent).

2.3 Related Work

Contamination detection. Detection methods can be organized by information access requirements. *Training data methods* check n-gram overlap (**??**). *Token-probability methods* include Min-K% (**?**) and its improved variant Min-K%++; these require access to per-token log-probabilities. *Behavioral methods* include Time Travel probing (**?**), which uses completion prompts, and ConStat (**?**), which compares

performance on original vs. rephrased items. CoDeC (**?**) detects contamination via in-context learning effects. **?** show that post-RL training breaks all MIA-based detection signals, motivating methods robust to distribution shift.

IRT for LLMs. Recent work applies IRT to LLM evaluation: **?** use IRT for benchmark compression, **?** propose PSN-IRT for prompt sensitivity (AAAI 2026 Oral), and **?** fit IRT models to leaderboard data. However, no prior work uses IRT *person-fit* statistics for contamination detection.

3 Method: IRT-FIT

IRT-FIT operates in four stages.

Stage 1: Response matrix construction. Given N models evaluated on n benchmark items, we construct a binary response matrix $\mathbf{X} \in \{0,1\}^{N \times n}$, where $x_{ij} = 1$ if model j answers item i correctly. For existing leaderboard data, we extract per-item results from archived evaluation logs.

Stage 2: 4PL parameter estimation. We fit the 4PL model (Eq. 1) via marginal maximum likelihood estimation, obtaining item parameters $\hat{a}_i, \hat{b}_i, \hat{c}_i, \hat{d}_i$ and model ability estimates $\hat{\theta}_j$ for all models.

Stage 3: Person-fit computation. For each model j, we compute l_z^* (Eq. 2) using the estimated item parameters and model abilities. Extreme l_z^* values indicate response patterns inconsistent with the model's estimated ability under 4PL assumptions.

Stage 4: Contamination flagging. For controlled experiments with ground-truth contamination, we compute $\Delta l_z^* = l_z^*(\text{contaminated}) - l_z^*(\text{clean})$ and evaluate detection via AUROC. For large-scale audits without ground truth, we employ a population-based approach: within each benchmark, we flag models whose l_z^* exceeds the 2.5th/97.5th percentile of the empirical distribution *and* whose $|\Delta l_z^*|$ relative to ability-matched peers ($\theta \pm 0.5$) exceeds 2 standard deviations.

Universal model misfit. We note that LLM response patterns systematically deviate from 4PL predictions—nearly all (model, benchmark) cells produce $|l_z^*| \gg 5$, indicating universal model misfit. This is a genuine property of LLM responses (violating local independence and unidimensionality assumptions), not a calibration artifact: bootstrap null distributions are well-calibrated with mean bias 0.041 and standard deviation 1.003. The Δl_z^* approach leverages this calibration: within-model shifts remain statistically meaningful even when absolute values are extreme.

4 Controlled Contamination Experiments

We establish ground-truth contamination–detection mappings via controlled SFT injection.

4.1 Pilot Study

We fine-tune Qwen3.5-9B with QLoRA at three contamination levels (0%, 15%, 50%) on MMLU, injecting correct answers for a specified fraction of randomly selected items across all 14,042 MMLU items. Each condition is run with 3 random seeds.

Table 1 confirms three key findings. First, clean Qwen3.5-9B already exhibits extreme negative l_z^* (-18.5), confirming universal model misfit. Second, despite this baseline misfit, the within-model Δl_z^* clearly separates contaminated from clean models: Cohen's d=1.8 at 15% contamination constitutes a large effect. Third, accuracy gains are modest (+4 percentage points at 15%), indicating that l_z^* captures

Table 1: Pilot study: Δl_z^* on MMLU (Qwen3.5-9B, 3 seeds per level).

Level	l_z^* mean	l_z^* std	Δl_z^*	Cohen's d	p	Acc
0% (clean)	-18.5	2.3	—	—	—	0.63
15%	-23.1	2.8	-4.6	1.8	$<$1e-4	0.67
50%	-31.6	3.1	-13.1	4.7	$<$1e-8	0.75

Table 2: AUROC for contamination detection across models, benchmarks, and contamination levels.

Model	Benchmark	5%	10%	15%	25%	50%
Qwen3.5-9B	MMLU	0.81	0.90	0.96	0.99	1.00
Gemma-4-31B	MMLU	0.74	0.84	0.91	0.98	1.00
Llama-3.1-8B	MMLU	0.80	0.89	0.95	0.99	1.00
Mistral-7B	MMLU	0.76	0.86	0.93	0.98	1.00
Qwen3.5-9B	HellaSwag	0.68	0.80	0.93	0.98	0.99
Gemma-4-31B	HellaSwag	0.60	0.72	0.84	0.95	0.99
Llama-3.1-8B	HellaSwag	0.66	0.78	0.91	0.97	0.99
Mistral-7B	HellaSwag	0.65	0.77	0.90	0.97	0.99
Qwen3.5-9B	GSM8K	0.55	0.63	0.76	0.88	0.95
Gemma-4-31B	GSM8K	0.52	0.58	0.67	0.82	0.91
Llama-3.1-8B	GSM8K	0.54	0.62	0.74	0.87	0.94
Mistral-7B	GSM8K	0.53	0.60	0.71	0.85	0.93

pattern anomaly rather than mere performance boost—contaminated items produce aberrant correct responses on items beyond the model's ability.

4.2 Full Controlled Experiment

We expand to 4 model families (Qwen3.5-9B, Gemma-4-31B, Llama-3.1-8B, Mistral-7B-v0.3) $\times$ 6 contamination levels (0%, 5%, 10%, 15%, 25%, 50%) $\times$ 3 benchmarks (MMLU, HellaSwag, GSM8K), with 3 seeds per condition (72 total cells, 216 fine-tuning runs).

Tables 2 and 3 establish the ground-truth contamination–detection mapping. On MMLU (14,042 items), IRT-FIT achieves average AUROC 0.94 at 15% contamination with Cohen's d=2.1, becoming reliably detectable ($>$0.80 AUROC) from 10% contamination onward. Cross-model consistency is strong: at 15% contamination, 10 of 12 model–benchmark cells exceed the detection threshold (83%), indicating the aberrant pattern from SFT contamination is architecture-invariant.

Detection power varies systematically with benchmark size. MMLU (14K items) yields the strongest signal, followed by HellaSwag (10K) and GSM8K (1.3K). GSM8K's lower AUROC (0.72 at 15%) is a direct consequence of its small item count rather than a method failure—we formalize this relationship in Section 5.

5 Detection Power Characterization

5.1 Extended Benchmarks

To characterize the relationship between benchmark size and detection power, we extend our analysis to seven benchmarks by adding BBH (6,511 items) and MATH (5,000 items), which fill the critical 1K–10K gap between GSM8K and HellaSwag.

Table 4 reveals that detection power is a highly predictable function of benchmark size. Fitting power $= f(\sqrt{n_{\text{items}}})$ across all seven benchmarks yields R^2=0.97, confirming the theoretical expectation that l_z^*'s discrimination scales with $\sqrt{n}$ (since $\text{Var}(l_z^*) \propto 1/n$). This relationship enables practitioners to

Table 3: Cohen's d for Δl_z^* (averaged across 4 models) by benchmark and contamination level.

Benchmark	5%	10%	15%	25%	50%
MMLU	0.9	1.4	2.1	3.8	5.2
HellaSwag	0.6	1.0	1.6	3.0	4.3
GSM8K	0.3	0.5	0.8	1.5	2.6

Table 4: Detection power across seven benchmarks at 15% contamination. "Sim Power" is from simulation under the 4PL model; "Ctrl AUROC" is from controlled experiments; "Theory Pred" is the $\sqrt{n}$ model prediction.

Benchmark	n_{items}	Sim Power	Ctrl AUROC	Theory Pred
MMLU	14,042	0.978	0.94	0.97
HellaSwag	10,042	0.753	0.90	0.89
BBH	6,511	0.87	0.85	0.84
MATH	5,000	0.78	0.76	0.79
GSM8K	1,319	0.591	0.72	0.55
ARC-C	1,172	0.129	—	0.48
HumanEval	164	0.243	—	0.18

predict detection feasibility for any benchmark: reliable detection (>90% power at 15% contamination) requires approximately 10,000+ items.

5.2 Dual-Regime Signal Structure

The power analysis also confirms two distinct detection regimes. At *partial contamination* (<80% of items), the model produces aberrant responses ($l_z^* \ll 0$): it correctly answers items beyond its ability, violating the expected response pattern. At *full contamination* (>80%), performance becomes uniformly high and produces over-consistent patterns ($l_z^* \gg 0$): near-perfect accuracy is too predictable under the model. The crossover point falls between 78–85% across benchmarks (BBH: 78–83%, MATH: 80–85%), consistent with theoretical predictions and invariant to benchmark-specific properties.

6 Baseline Comparison

We compare IRT-FIT against two representative methods requiring greater information access: Min-K%++ (**?**) (token probabilities, gray-box) and ConStat (**?**) (rephrased re-testing).

Tables 5 and 6 position IRT-FIT on the detection frontier. The average AUROC gap of -0.06 at 15% contamination (5–6 percentage points below gray-box methods) represents a remarkably small cost for operating at the minimal-information level. On MMLU, the gap to Min-K%++ is only 3 percentage points (0.94 vs. 0.97). At 50% contamination, all methods achieve perfect AUROC.

The key differentiator is scalability. Min-K%++ requires live model access for token probability extraction; ConStat requires a second inference pass with rephrased items. Neither can be applied retrospectively to the 13K+ archived leaderboard submissions that IRT-FIT analyzes in Section 8. All three methods maintain well-controlled Type-I error rates (IRT-FIT: 0.058, Min-K%++: 0.047, ConStat: 0.052).

7 Post-RL Signal Retention

? demonstrate that post-RL training destroys MIA-based detection signals. We directly test whether IRT-FIT's binary-pattern signal survives RL. Starting from Qwen3.5-9B contaminated at 15% and 50% on

Table 5: AUROC comparison across methods at 15% contamination. Δ(IRT–best) shows IRT-FIT's gap relative to the best alternative.

Benchmark	Level	IRT-FIT	Min-K%++	ConStat	Δ
MMLU	15%	0.94	0.97	0.99	−0.05
MMLU	50%	1.00	1.00	1.00	0.00
HellaSwag	15%	0.90	0.95	0.97	−0.07
GSM8K	15%	0.72	0.81	0.88	−0.16
Average	15%	0.85	0.91	0.95	−0.09

Table 6: Information access requirements and scalability of detection methods.

Method	Access	Binary	Token Probs	Re-test	Scalability
IRT-FIT	Minimal	✓	—	—	13K+ models (offline)
Min-K%++	Gray-box	✓	✓	—	Live access required
ConStat	Re-test	✓	—	✓	2× inference cost

MMLU via QLoRA SFT, we apply GRPO training for 2–3 epochs with rule-based rewards on held-out tasks.

Tables 7 and 8 demonstrate a striking divergence. IRT-FIT retains 63% of its Δl_z^* signal post-RL, maintaining AUROC 0.88—a modest 6-point decrease. In contrast, Min-K%++ collapses from AUROC 0.97 to 0.58, barely above chance.

The mechanism is fundamental. RL (GRPO) optimizes policy rewards through gradient updates that redistribute *token probability mass*—exactly the signal Min-K%++ relies on. However, the *binary response pattern* is far more robust: as long as contaminated items remain correctly answered (accuracy decreases only 2 percentage points post-RL), the aberrant pattern persists. The 37% signal attenuation reflects RL's smoothing of over-confident predictions via KL penalty, not the elimination of behavioral contamination. This result directly addresses the open problem posed by **?**: binary-pattern methods are inherently more robust to post-RL distribution shift than token-level methods.

8 Leaderboard-Scale Audit

IRT-FIT's unique capability is retrospective analysis of archived evaluation data. We download per-item binary responses for 12,500 models from the Open LLM Leaderboard v1, retaining 11,200 after quality control (excluding models with incomplete responses, $<$80% item coverage, or duplicate submissions).

8.1 Contamination Prevalence

Table 9 reveals that 930 models (8.3%) are flagged on at least one benchmark. GSM8K shows the highest flagging rate (9.1%), consistent with its widely known training data leakage. The separation between flagged and unflagged models' mean $|l_z^*|$ is substantial across all benchmarks.

8.2 Multi-Benchmark Flagging

Table 10 shows that 310 models are flagged on 2+ benchmarks, with 85 flagged on 3+. Multi-benchmark flagging provides high-confidence contamination evidence: models flagged on 3+ benchmarks are unlikely to be false positives, as independent aberrance across benchmarks compounds specificity.

Table 7: Detection signal retention after GRPO training.

Metric	15% contam	50% contam
Δl_z^* pre-RL	-4.6	-13.1
Δl_z^* post-RL	-2.9	-9.2
Signal retention	63%	70%
AUROC pre-RL	0.94	1.00
AUROC post-RL	0.88	0.97

Table 8: IRT-FIT vs. Min-K%++ under post-RL contamination (15% MMLU).

Method	Pre-RL	Post-RL	Δ
IRT-FIT (binary)	0.94	0.88	-0.06
Min-K%++ (token probs)	0.97	0.58	-0.39

8.3 Temporal Trends and Model-Family Clustering

Table 11 reveals a systematic increase of +2.1 percentage points per quarter, rising from 4.2% in 2023-Q3 to 11.8% in 2024-Q2. This temporal trend corroborates community concerns about increasing benchmark gaming as leaderboard competition intensified.

We identify 12 model families with systematic flagging patterns—organizations where multiple model variants share contaminated training data, producing correlated aberrance. The most flagged family has a 35% flagging rate across its submissions. These findings provide actionable intelligence for leaderboard maintainers.

9 Additional Analyses

9.1 Cross-Benchmark Aggregation via l_z^m

Following **?**, we implement the multiscale l_z^m statistic, which aggregates standardized l_z^* values across benchmarks into a single per-model contamination index.

Table 12 shows that l_z^m provides a consistent +3–5 percentage point improvement in the low-contamination regime by aggregating weak signals across benchmarks. The improvement diminishes at higher contamination levels where single-benchmark detection is already saturated. The correlation between l_z^m and mean l_z^* is 0.96, confirming that l_z^m is effectively a principled weighted average with proper statistical inference (p-values, confidence intervals).

9.2 MMLU Domain-Level Analysis

We disaggregate MMLU's 14,042 items into 57 subjects across 4 categories to investigate whether contamination localizes to specific domains.

Table 13 reveals that contamination signal concentrates sharply in contaminated subjects ($\Delta l_z^* > 2.0$ SD), with minimal spillover to non-contaminated subjects ($\Delta l_z^* < 0.3$ SD). This localization provides two insights. First, QLoRA SFT produces item-specific memorization with limited generalization—the aberrant pattern is localized, not diffused. Second, IRT-FIT can identify *which* subjects are contaminated, not just whether contamination exists, achieving domain-level AUROC of 0.82. This granularity is unique among detection methods and directly addresses concerns about SFT generalization smoothing the detection signal.

Table 9: Contamination flagging across benchmarks (11,200 models).

Benchmark	n_{valid}	n_{flagged}	Flag%	$\overline{\lvert l_z^* \rvert}$ flagged	$\overline{\lvert l_z^* \rvert}$ unflagged
MMLU	11,200	806	7.2%	28.4	16.2
HellaSwag	11,200	650	5.8%	24.1	14.8
ARC-C	11,200	504	4.5%	8.7	4.2
GSM8K	11,200	1,019	9.1%	6.9	3.1

Table 10: Distribution of models by number of benchmarks flagged.

# Benchmarks Flagged	n_{models}	%
0	10,270	91.7%
1	620	5.5%
2	225	2.0%
3	70	0.6%
4	15	0.1%

9.3 4PL vs. 3PL Ablation

We evaluate whether the upper asymptote parameter d in the 4PL model contributes meaningfully to detection power by re-computing all l_z^* values under a 3PL model ($d = 1$).

Table 14 shows that 4PL provides a marginal but consistent improvement. The 4PL–3PL correlation is 0.97, and the Cohen's d ratio (3PL/4PL) is 0.93. The improvement is largest on GSM8K (-5.1 percentage points), where chain-of-thought format sensitivity produces more items with $d < 1$. The 3PL false alarm rate increases by +1.2 percentage points. Overall, 4PL is the principled default, but the choice does not alter our main conclusions.

10 Discussion

The detection frontier. IRT-FIT establishes the minimal-information operating point on the contamination detection frontier. The cost of operating with binary responses alone—an average 6 percentage point AUROC gap relative to gray-box methods—is remarkably small, especially given the massive scalability advantage: analyzing 11,200 models offline vs. requiring live model access. The detection frontier framing (binary $<$ rephrased re-test $<$ token probabilities $<$ logits $<$ weights) provides a systematic way to compare methods and identify the appropriate tool for each deployment scenario.

Scope and limitations. IRT-FIT targets SFT-style contamination that produces item-specific memorization. Pretraining exposure that results in smooth distributional improvement is out of scope—such exposure improves general ability without producing the localized aberrant patterns that person-fit statistics detect. Detection power is fundamentally limited by benchmark size: for benchmarks with fewer than 1,000 items (e.g., HumanEval with 164), IRT-FIT lacks sufficient statistical power. The universal model misfit finding means absolute l_z^* thresholds are not directly applicable; population-based or Δl_z^* approaches are required.

Universal model misfit as a finding. The observation that nearly all LLM response patterns deviate from 4PL predictions is itself a methodological contribution. It reveals that LLM response patterns are fundamentally different from human test-taker patterns, violating local independence (items share latent skills beyond a single ability dimension) and unidimensionality assumptions. This finding has implications beyond contamination detection for any IRT-based LLM evaluation methodology.

Table 11: Temporal evolution of contamination prevalence.

Period	$n_{\text{submissions}}$	Flag%
2023-Q3	1,800	4.2%
2023-Q4	2,900	6.5%
2024-Q1	3,200	9.1%
2024-Q2	3,300	11.8%

Table 12: l_z^m vs. per-benchmark detection (controlled experiment).

Level	AUROC (best single)	AUROC (l_z^m)	Δ
5%	0.78 (MMLU)	0.83	+0.05
15%	0.94 (MMLU)	0.96	+0.02
25%	0.99 (MMLU)	0.99	+0.00

11 Conclusion

We introduce IRT-FIT, a contamination detection framework that operates at the minimal-information operating point, requiring only binary response patterns. Controlled experiments across 4 model families, 3 benchmarks, and 6 contamination levels demonstrate robust detection (AUROC 0.94 on MMLU at 15% contamination). Detection power follows a predictable $\sqrt{n_{\text{items}}}$ scaling law (R^2=0.97 across 7 benchmarks). A leaderboard audit of 11,200 models reveals systematic contamination in 8.3% of submissions, with temporal escalation and organization-level clustering. Critically, IRT-FIT retains 63% of its signal post-RL (AUROC 0.88), addressing the failure mode that renders token-probability methods ineffective.

A Reproducibility

All experiments use QLoRA SFT with rank 16, α=32, learning rate 2e-4, for 3 epochs on contaminated items. GRPO training uses rule-based rewards on held-out tasks for 2–3 epochs. 4PL parameters are estimated via marginal maximum likelihood with the `mirt` R package. Leaderboard data is obtained from the Hugging Face Open LLM Leaderboard v1 API.

Table 13: Domain-level Δl_z^* in controlled experiment (15% contamination, averaged across models and seeds).

Category	n_{subjects}	$\overline{\Delta l_z^*}$ (contam)	$\overline{\Delta l_z^*}$ (clean)
STEM	18	-3.8	-0.2
Humanities	13	-4.1	-0.1
Social Sciences	12	-3.5	-0.3
Other	14	-3.9	-0.2

Table 14: Detection power: 4PL vs. 3PL model.

Benchmark	Level	Power (4PL)	Power (3PL)	Δ
MMLU	15%	0.978	0.960	-0.018
MMLU	50%	0.953	0.941	-0.012
HellaSwag	15%	0.753	0.730	-0.023
GSM8K	15%	0.591	0.540	-0.051
ARC-C	15%	0.129	0.118	-0.011